DAVID M. KROENKE and DAVID J. AUER

DATABASE CONCEPTS, 6th Edition

Chapter Four

Data Modeling and the Entity-Relationship Model

Chapter Objectives

- Learn the basic stages of database development
- Understand the purpose and role of a data model
- Know the principal components of the E-R data model
- Understand how to interpret traditional E-R diagrams
- Understand how to interpret the Information Engineering (IE) model's Crow's Foot E-R diagrams
- Learn to construct E-R diagrams
- Know how to represent 1:1, 1:N, N:M, and binary relationships with the E-R model

Chapter Objectives
(Cont'd)

- Understand two types of weak entities and know how to use them
- Understand nonidentifying and identifying relationships and know how to use them
- Know how to represent subtype entities with the E-R model
- Know how to represent recursive relationships with the E-R model
- Learn how to create an E-R diagram from source documents

Three Stages of Database Development

- The three stages of database development are:
  - Requirements Analysis Stage
  - Component Design Stage
  - Implementation Stage
- These three stages are part of the five stage Systems Development Life Cycle (SDLC) model

The Requirements Analysis Stage

- Sources of requirements
  - User Interviews
  - Forms
  - Reports
  - Queries
  - Use Cases
  - Business Rules

Requirements Become the E-R Data Model

- After the requirements have been gathered, they are transformed into an Entity Relationship (E-R) Data Model.
- The most important elements of E-R Models are:
  - Entities
  - Attributes
  - Identifiers
  - Relationships

Entity Class versus Entity Instance

- An **entity class** is a description of the structure and format of the occurrences of the entity.
- An **entity instance** is a specific occurrence of an entity within an entity class.

Entity Class and Entity Instance

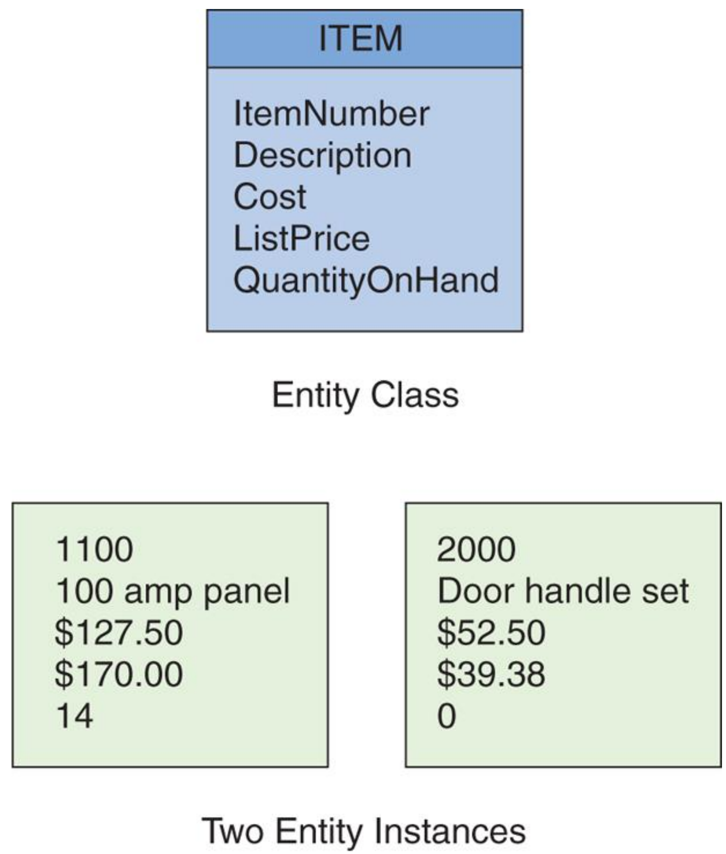


Figure 4-2: The ITEM Entity and Two Entity Instances

Attributes

- Entities have **attributes** that describe the entity's characteristics:
  - ProjectName
  - StartDate
  - ProjectType
  - ProjectDescription
- Attributes have a data type and properties.

Identifiers

- Entity instances have **identifiers.**
- An identifier will identify a particular instance in the entity class:
  - SocialSecurityNumber
  - StudentID
  - EmployeeID

Identifier Types

- Uniqueness
  - Identifiers may be **unique** or **nonunique.**
  - If the identifier is unique, the data value for the identifier must be unique for all instances.
- Composite
  - A **composite identifier** consists of two or more attributes.
    - E.g., OrderNumber & LineItemNumber are both required.

Levels of Entity Attribute Display

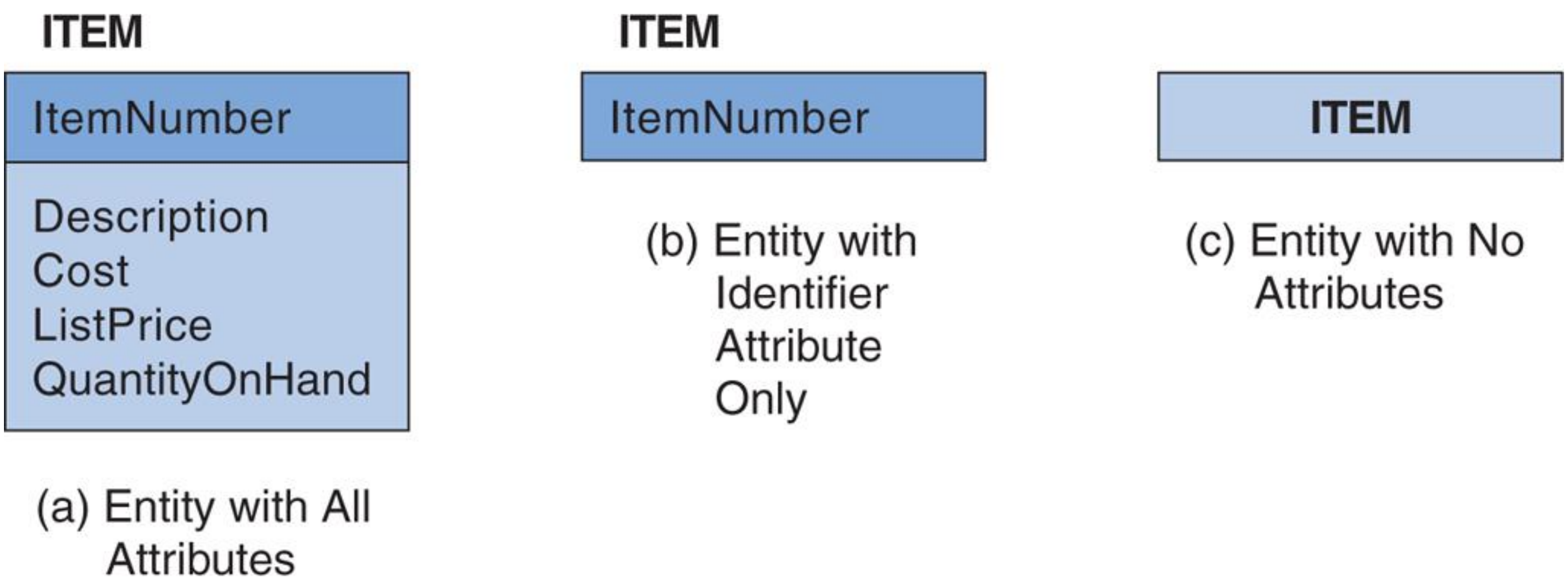
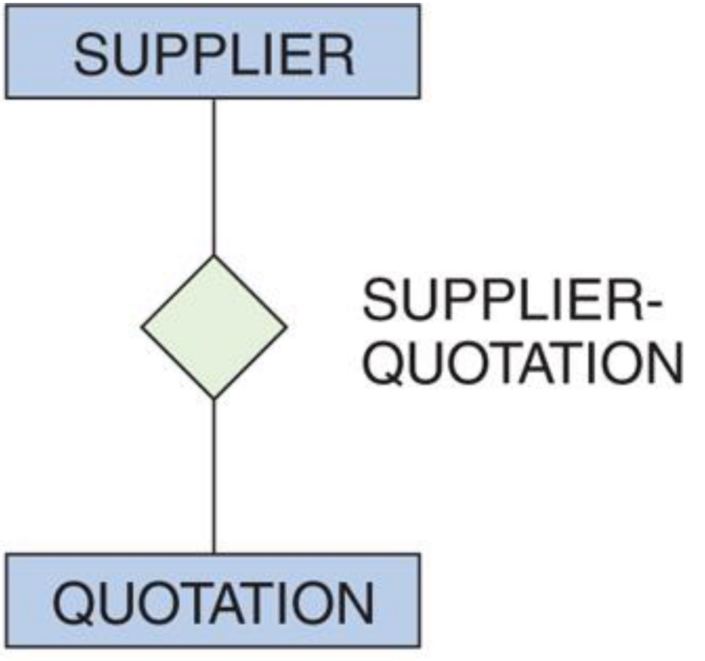


Figure 4-3: Levels of Entity Attribute Display

Relationships

- Entities can be associated with one another in **relationships.**
- Relationship *degree* defines the number of entity classes participating in the relationship:
  - Degree 2 is a **binary relationship.**
  - Degree 3 is a **ternary relationship.**

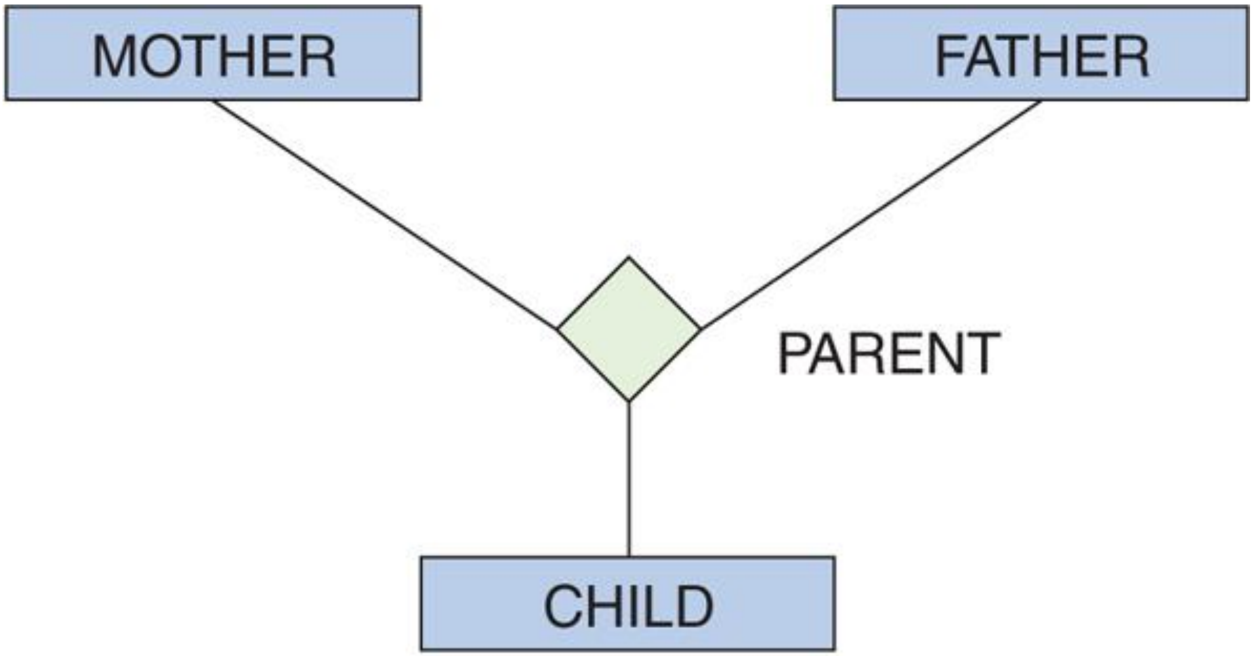
Degree 2 Relationship: Binary



(a) Binary Relationship

Figure 4-4: Example Relationships

Degree 3 Relationship: Ternary



(b) Ternary Relationship

Figure 4-4: Example Relationships

One-to-One Binary Relationship

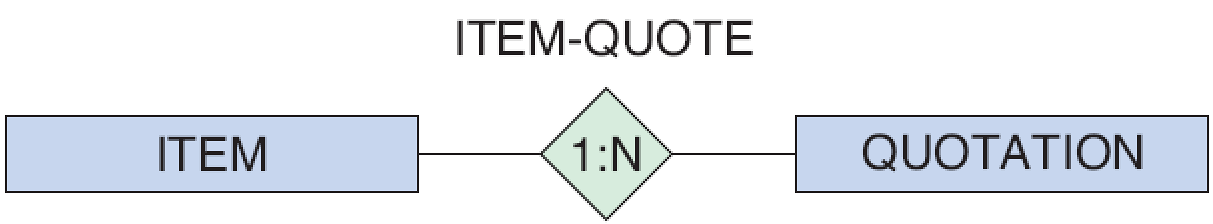
- **1:1 (one-to-one)**
  - A single entity instance in one entity class is related to a single entity instance in another entity class.
    - An employee may have no more than one locker; and
    - A locker may only be accessible by one employee

(a) One-to-One Relationship

Figure 4-5: Three Types of Binary Relationships

One-to-Many Binary Relationship

- **1:N (one-to-many)**
  - A single entity instance in one entity class is related to many entity instances in another entity class.
    - A quotation is associated with only one item; and
    - An item may have several quotations

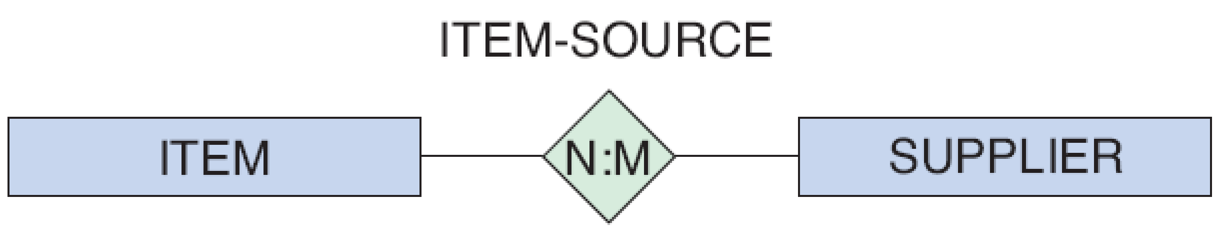


(b) One-to-Many Relationship

Figure 4-4: Three Types of Binary Relationships

Many-to-Many Binary Relationship

- **N:M (many-to-many)**
  - Many entity instances in one entity class is related to many entity instances in another entity class:
    - a supplier may supply several items; and
    - a particular item may be supplied by several suppliers.



(c) Many-to-Many Relationship

Figure 4-5: Three Types of Binary Relationships

Maximum Cardinality

- Relationships are named and classified by their **cardinality**, which is a word that means *count.*
- Each of the three types of binary relationships shown above have different *maximum cardinalities.*
- **Maximum cardinality** is the maximum number of entity instances that may participate in a relationship instance—one, many, or some other fixed number.

Minimum Cardinality

- **Minimum cardinality** is the minimum number of entity instances that *must* participate in a relationship instance.
- These values typically assume a value of zero (optional) or one (mandatory).

Cardinality Example

- Maximum cardinality is many for both ITEM and SUPPLIER.
- Minimum cardinality is zero (optional) for ITEM and one (mandatory) SUPPLIER.
  - A SUPPLIER does not have to supply an ITEM.
  - An ITEM must have a SUPPLIER.

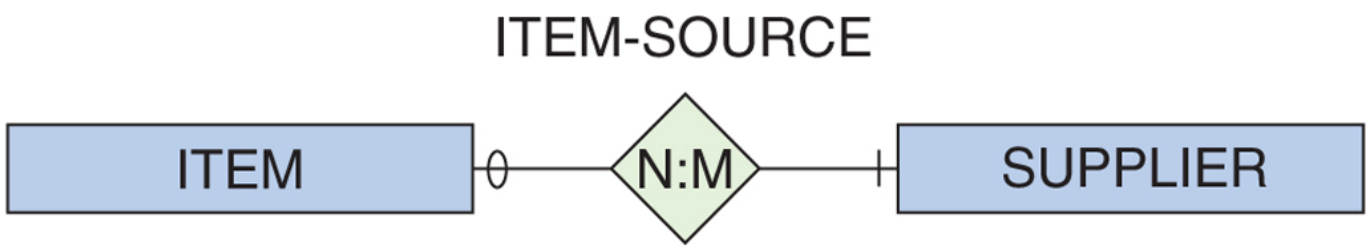


Figure 4-6: A Relationship with Minimum Cardinalities

Entity-Relationship Diagrams

- The diagrams in previous slides are called **entity-relationship diagrams.**
  - Entity classes are shown by rectangles.
  - Relationships are shown by diamonds.
  - The maximum cardinality of the relationship is shown inside the diamond.
  - The minimum cardinality is shown by the oval or hash mark next to the entity.
  - The name of the entity is shown inside the rectangle.
  - The name of the relationship is shown near the diamond.

HAS-A Relationships

- The relationships in the previous slides are called **HAS-A relationships.**
- The term is used because each entity instance *has a* relationship to a second entity instance:
  – An employee has a badge.
  – A badge has an employee.

Types of Entity-Relationship Diagrams

- **Information Engineering (IE)** [James Martin 1990]—Uses “crow’s feet” to show the many sides of a relationship, and it is sometimes called the **crow’s foot model**.
- **Integrated Definition 1, Extended** 3 **(IDEF1X)** is a version of the E-R model that is a national standard.
- **Unified Modeling Language (UML)** is a set of structures and techniques for modeling and designing object-oriented programs (OOP) and applications

Crow's Foot Example: One-to-Many Relationship

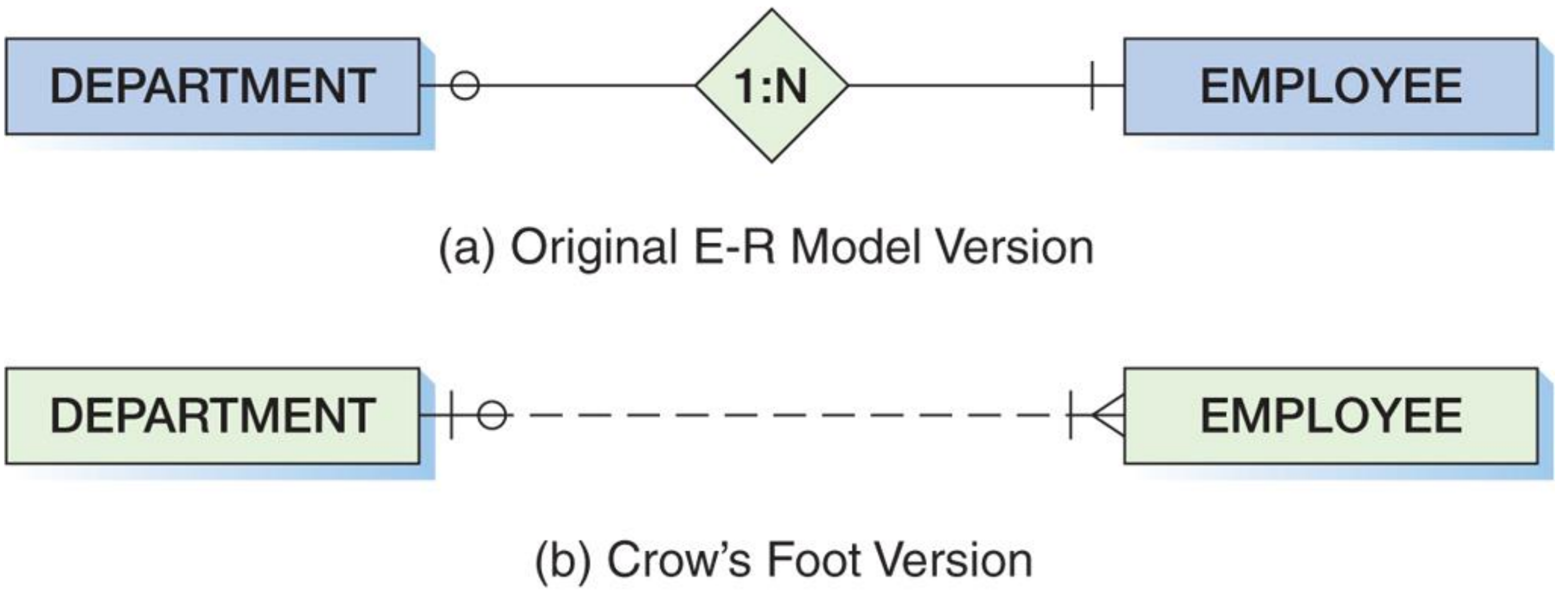


Figure 4-7: Two Versions of a 1:N Relationship

Crow's Foot Symbols

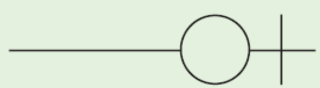
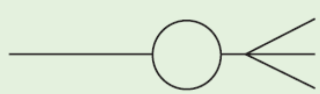
Symbol	Meaning	Numeric Meaning
	Mandatory—One	Exactly one
	Mandatory—Many	One or more
	Optional—One	Zero or one
	Optional—Many	Zero or more

Figure 4-8: Crow's Foot Notation

Crow's Foot Example: Many-to-Many Relationship

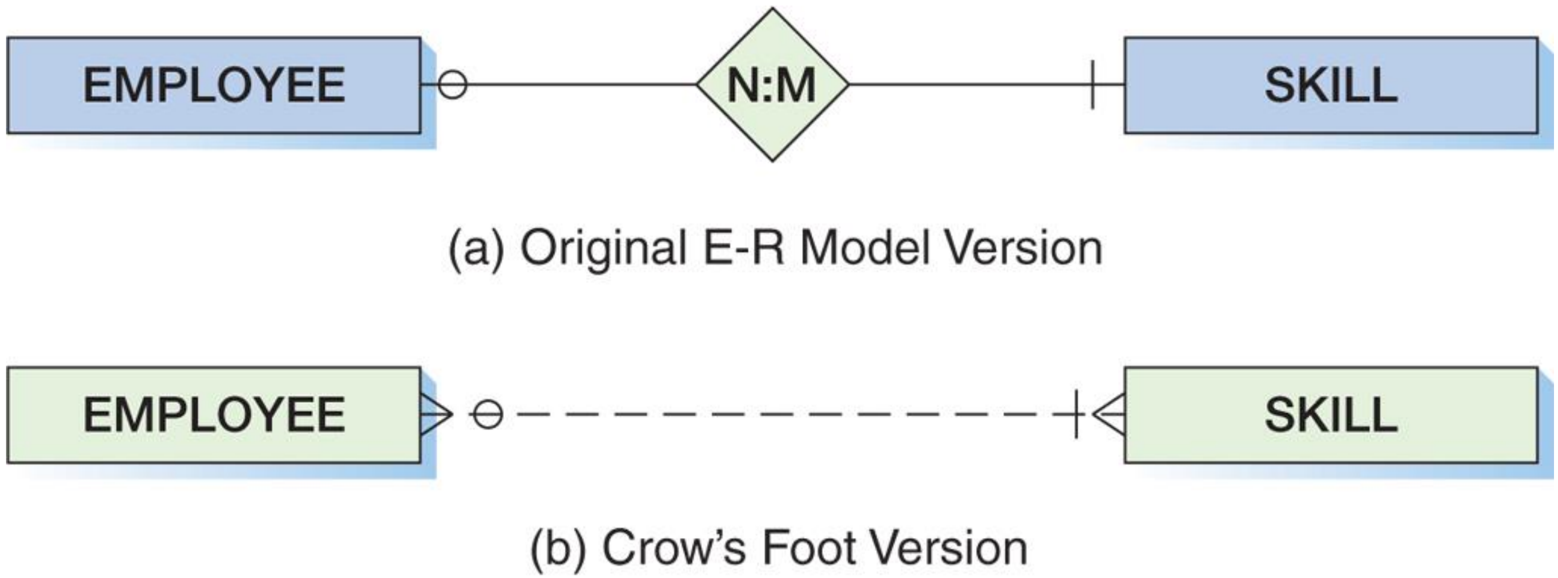


Figure 4-9: Two Versions of an N:M Relationship

Weak Entity

- A **weak entity** is an entity that cannot exist in the database without the existence of another entity.
- Any entity that is not a weak entity is called a **strong entity**.

ID-Dependent Weak Entities

- An ID-Dependent weak entity is a weak entity that cannot exist without its parent entity.
- An ID-dependent weak entity has a composite identifier.
  - The first part of the identifier is the identifier for the strong entity.
  - The second part of the identifier is the identifier for the weak entity itself.

ID-Dependent Weak Entity Examples

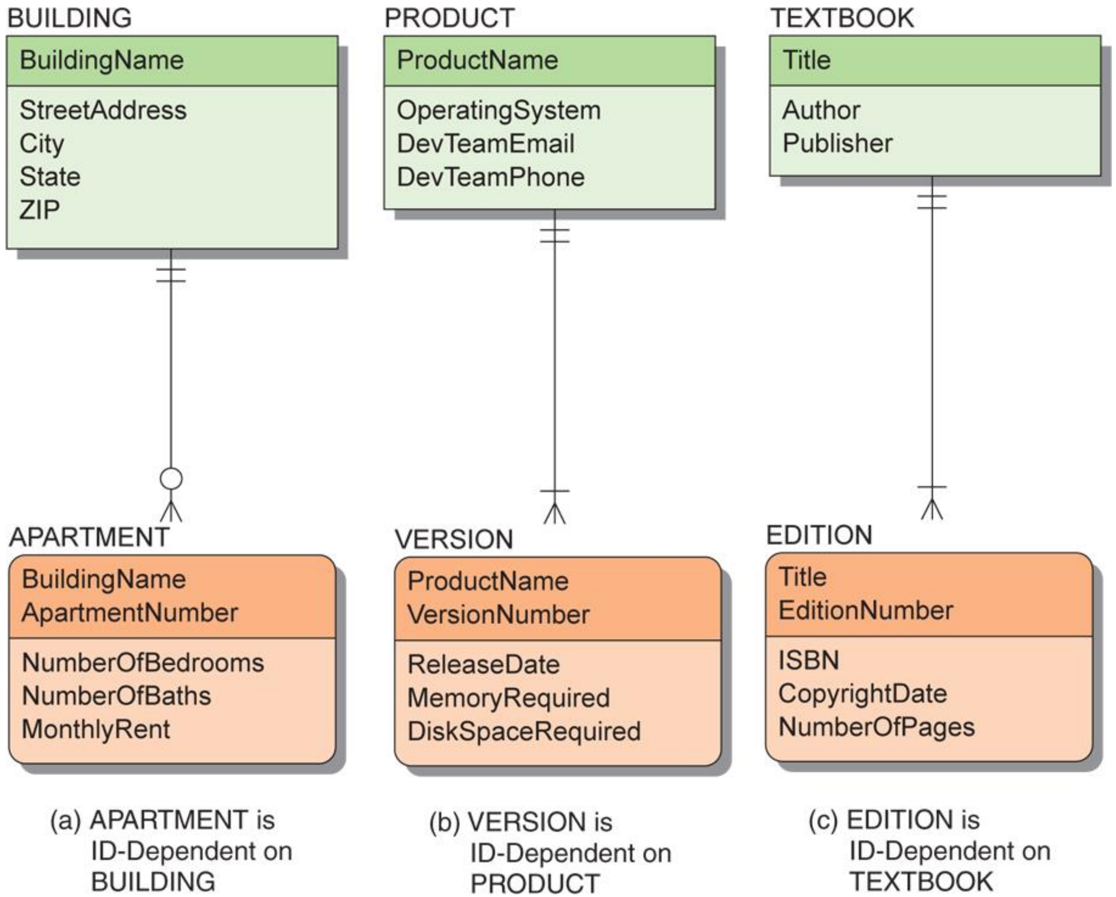


Figure 4-10: Example ID-Dependent Entities

Weak Entity Relationships

- The relationship between a strong and weak entity is termed an **identifying relationship** if the weak entity is ID-dependent.
  - Represented by a *solid line*
- The relationship between a strong and weak entity is termed a **nonidentifying relationship** if the weak entity is non-ID-dependent.
  - Represented by a *dashed line*
  - Also used between strong entities

Weak Entity Identifier: Non-ID-dependent

- All ID-dependent entities are weak entities, but there are other entities that are weak but not ID-dependent.
- A non-ID-dependent weak entity may have a single or composite identifier, but the identifier of the parent entity will be a *foreign key.*

Non-ID-Dependent Weak Entity Examples

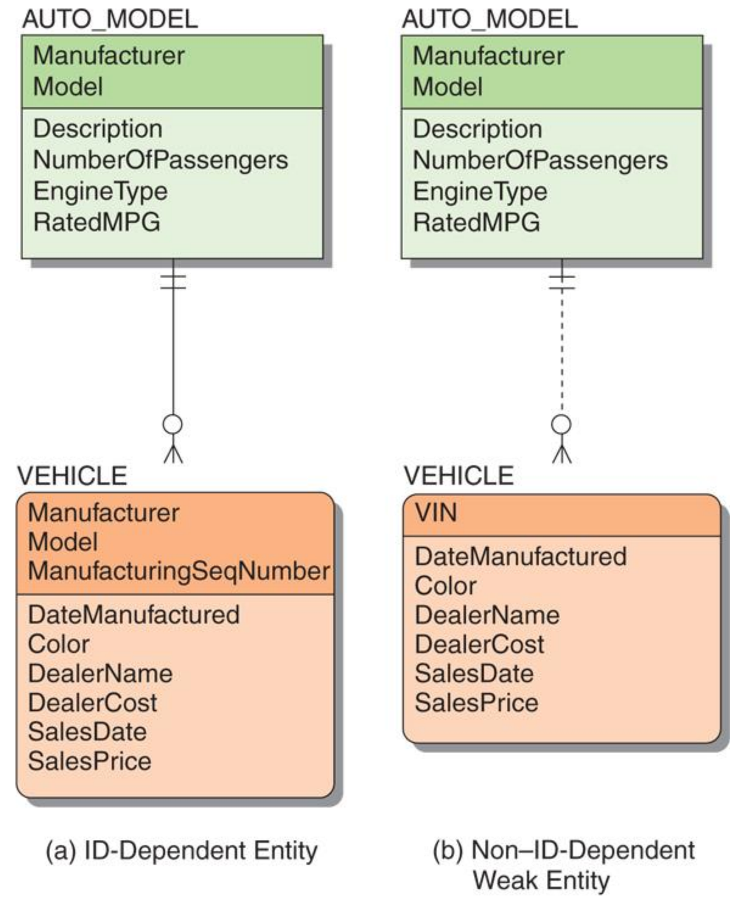


Figure 4-11: Weak Entity Examples

Strong and Weak Entity Examples

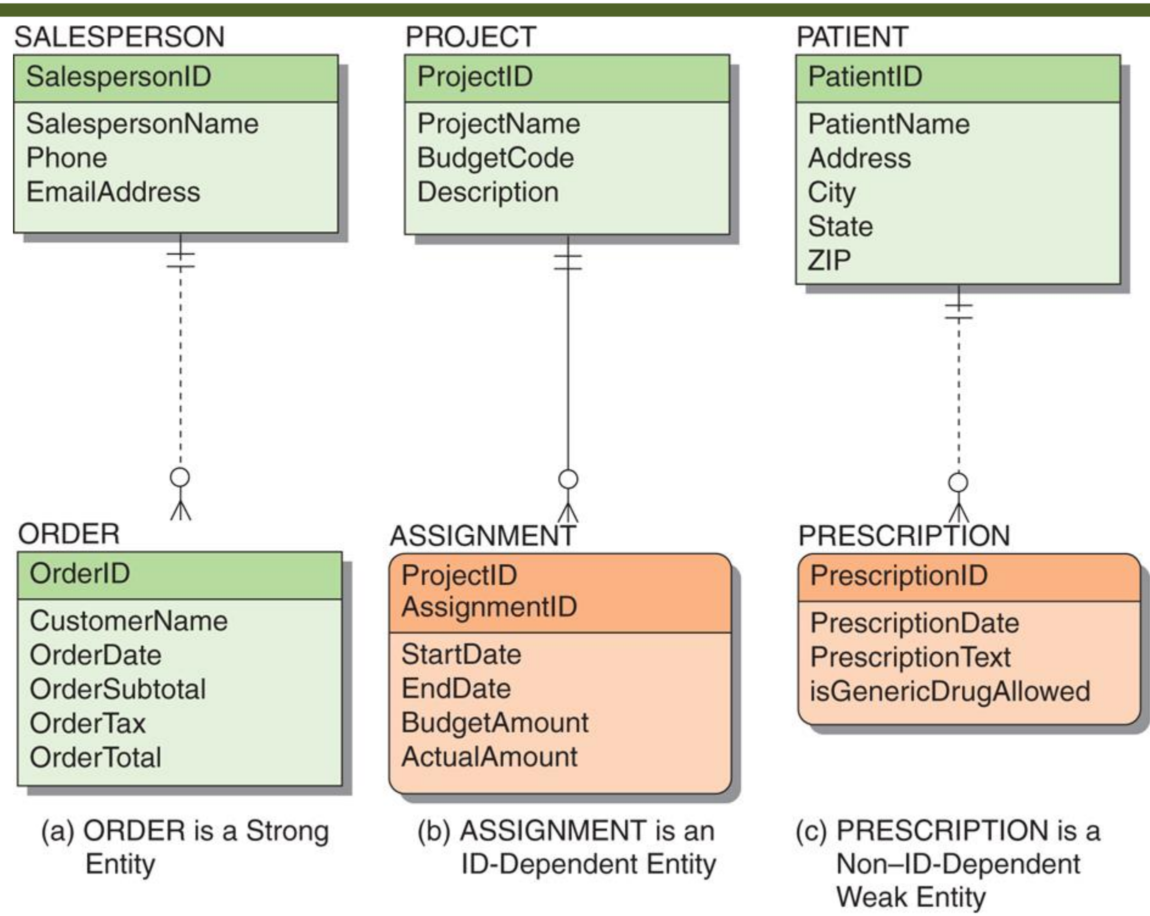


Figure 4-12: Examples of Required Entities

Associative Entities

- An **associative** entity (also called an **association** entity) is used when there are attributes that are associated with the relationship between two entities rather than with either of the two entities themselves.
- A new entity is then created to:
  - Link the two original entities
  - Hold the attributes

Associative Entities (Cont'd)

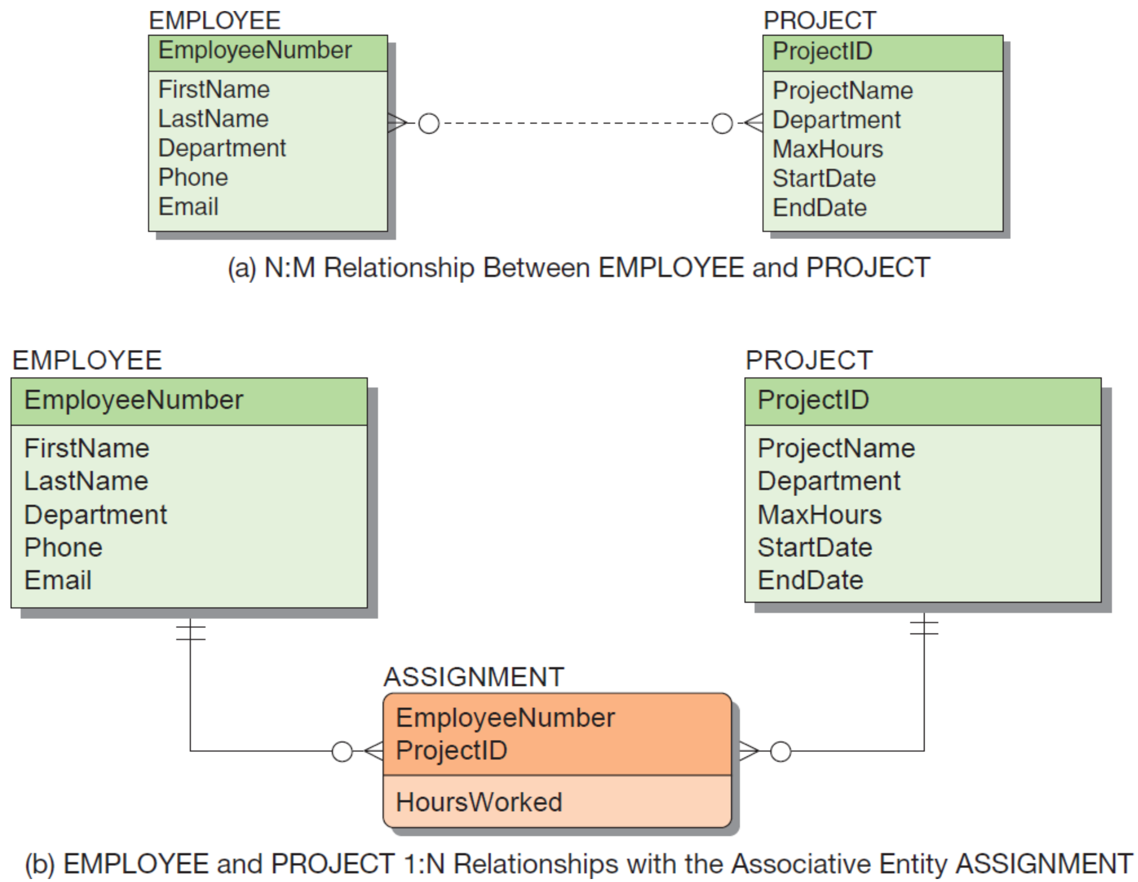


Figure 4-13: The Associative Entity

Subtype Entities

- A **subtype** entity is a special case of another entity called **supertype.**
- An attribute of the supertype may be included that indicates which of the subtypes is appropriate for a given instance; this attribute is called a **discriminator.**
- Subtypes can be *exclusive* or *inclusive.*
  - If **exclusive**, the supertype relates to at most one subtype.
  - If **inclusive**, the supertype can relate to one or more subtypes.

Subtype Entity Identifiers

- The relationships that connect supertypes and subtypes are called **IS-A relationships** because a subtype is the same entity as the supertype.
- The identifier of a supertype and all of its subtypes is the same attribute.

Subtype Entity Examples

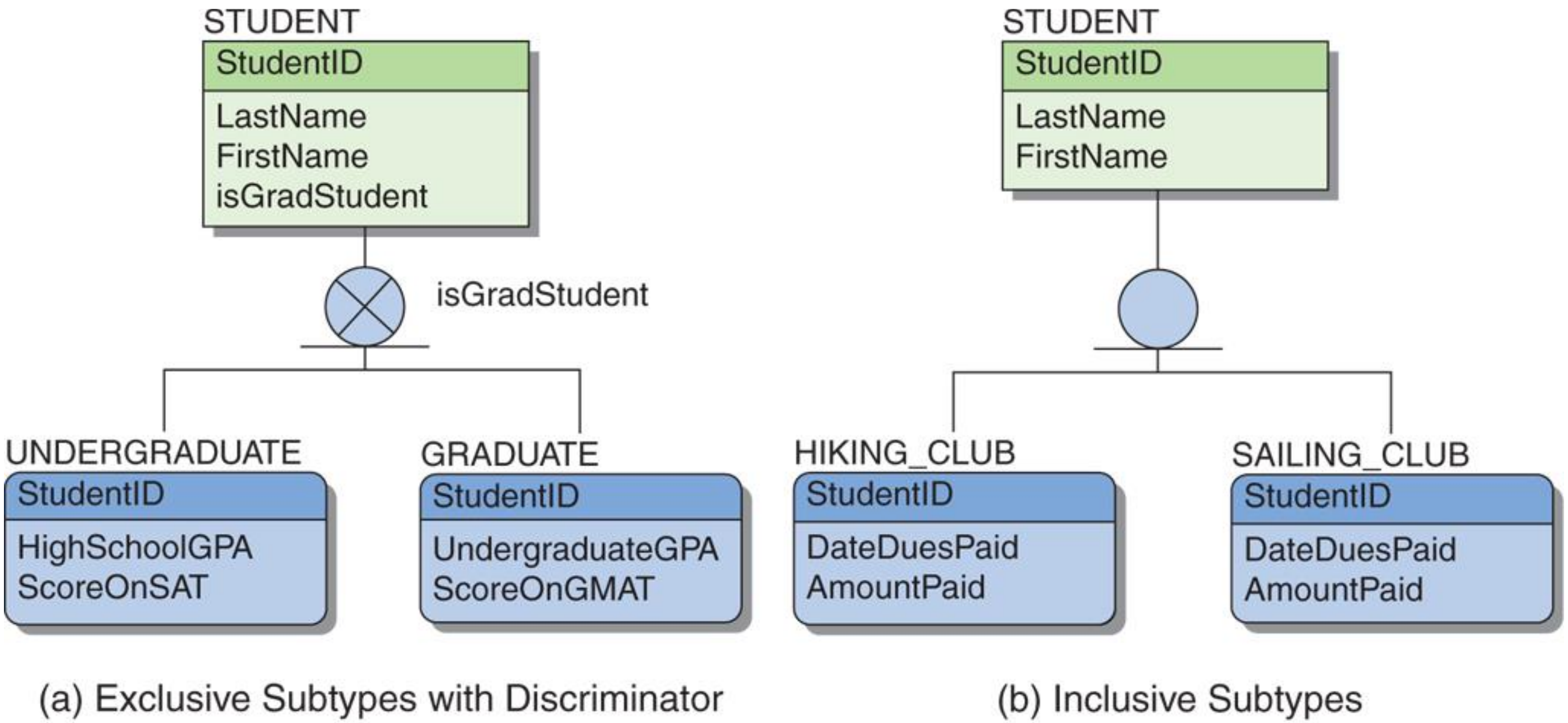


Figure 4-14: Example Subtype Entities

Recursive Relationships

- It is possible for an entity to have a relationship to itself—this is called a **recursive relationship.**

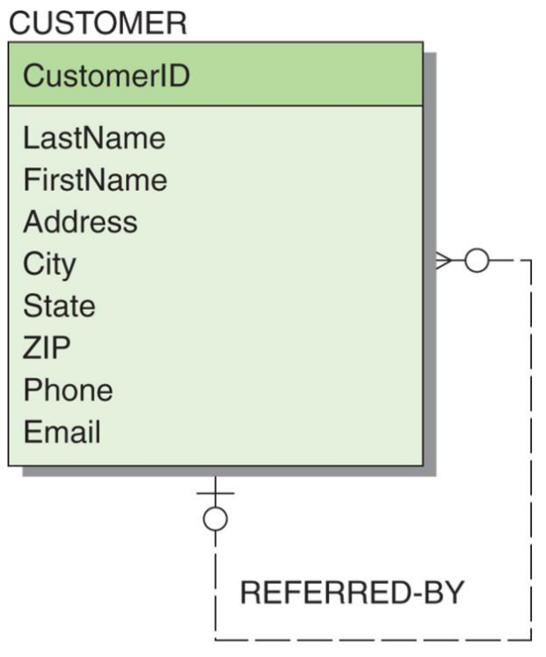


Figure 4-15:
Example Recursive Relationship

Developing an E-R Diagram

- Heather Sweeney Designs will be used as an ongoing example throughout Chapters 4, 5, 6 and 7.
  - Heather Sweeney is an interior designer who specializes in home kitchen design.
  - She offers a variety of free seminars at home shows, kitchen and appliance stores and other public locations.
  - She earns revenue by selling books and videos that instruct people on kitchen design.
  - She also offers custom-design consulting services.

Heather Sweeney Designs: The Seminar Customer List

Heather Sweeney Designs
Seminar Customer List

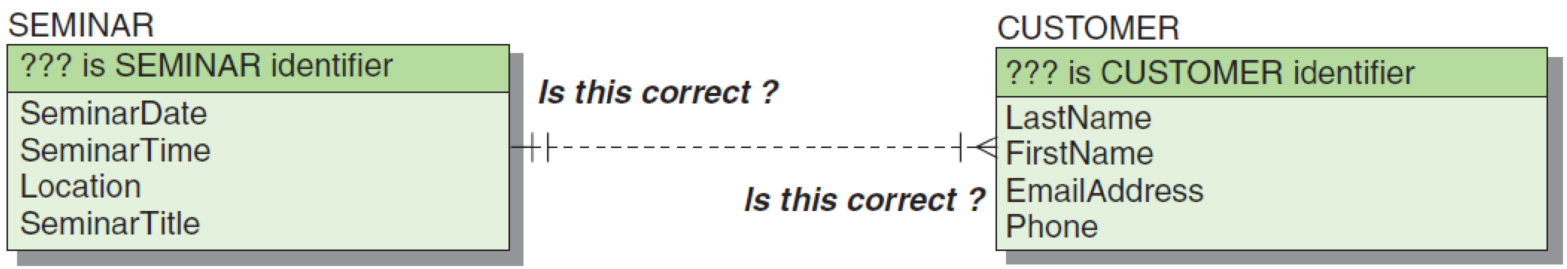
Date: October 11, 2014 Location: San Antonio Convention Center

Time: 11 AM Title: Kitchen on a Budget

Name	Phone	Email Address
Nancy Jacobs	817–871–8123	NJ@somewhere.com
Chantel Jacobs	817–871–8234	CJ@somewhere.com
Ralph Able	210–281–7687	RA@somewhere.com
Etc.		
27 names in all		

Figure 4-16: Example Seminar Customer List

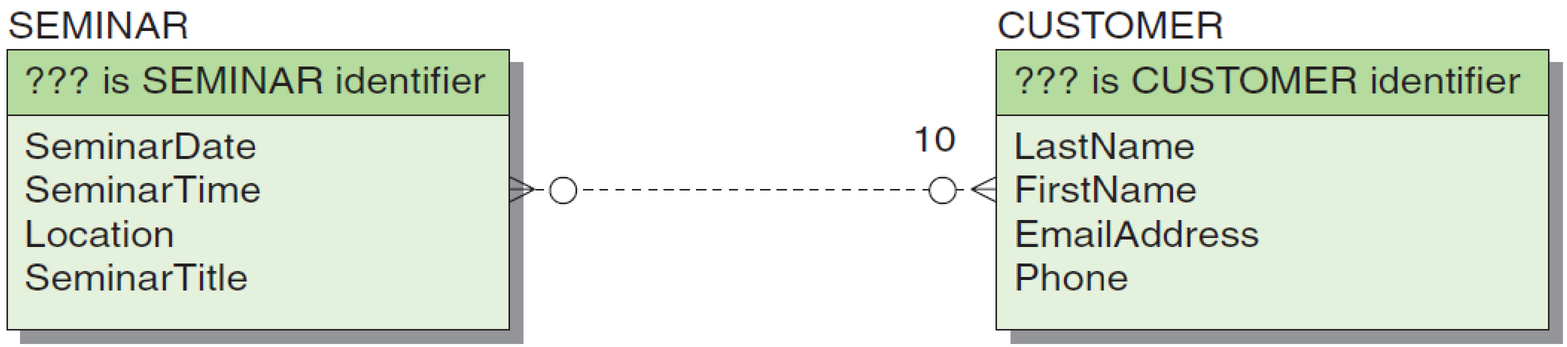
Heather Sweeney Designs: Initial E-R Diagram I



(a) First Version of the SEMINAR and CUSTOMER E-R Diagram

Figure 4-17: Initial E-R Diagram for Heather Sweeney Designs

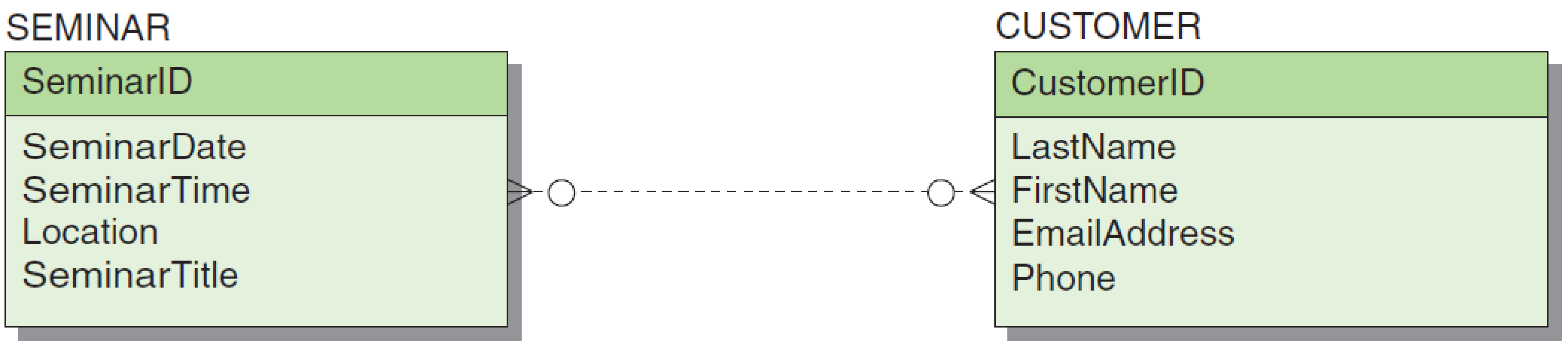
Heather Sweeney Designs: Initial E-R Diagram II



(b) Second Version of the SEMINAR and CUSTOMER E-R Diagram

Figure 4-17: Initial E-R Diagram for Heather Sweeney Designs

Heather Sweeney Designs: Initial E-R Diagram III



(c) Third Version of the SEMINAR and CUSTOMER E-R Diagram

Figure 4-17: Initial E-R Diagram for Heather Sweeney Designs

Heather Sweeney Designs: The Customer Form Letter

Heather Sweeney Designs
122450 Rockaway Road
Dallas, Texas 75227
972-233-6165

Ms. Nancy Jacobs
1400 West Palm Drive
Fort Worth, Texas 76110

Dear Ms. Jacobs:

Thank you for attending my seminar “Kitchen on a Budget” at the San Antonio Convention Center. I hope that you found the seminar topic interesting and helpful for your design projects.

As a seminar attendee, you are entitled to a 15 percent discount on all of my video and book products. I am enclosing a product catalog and I would also like to invite you to visit our Web site at www.Sweeney.com.

Also, as I mentioned at the seminar, I do provide customized design services to help you create that just-perfect kitchen. In fact, I have a number of clients in the Fort Worth area. Just give me a call at my personal phone number of 555-122-4873 if you’d like to schedule an appointment.

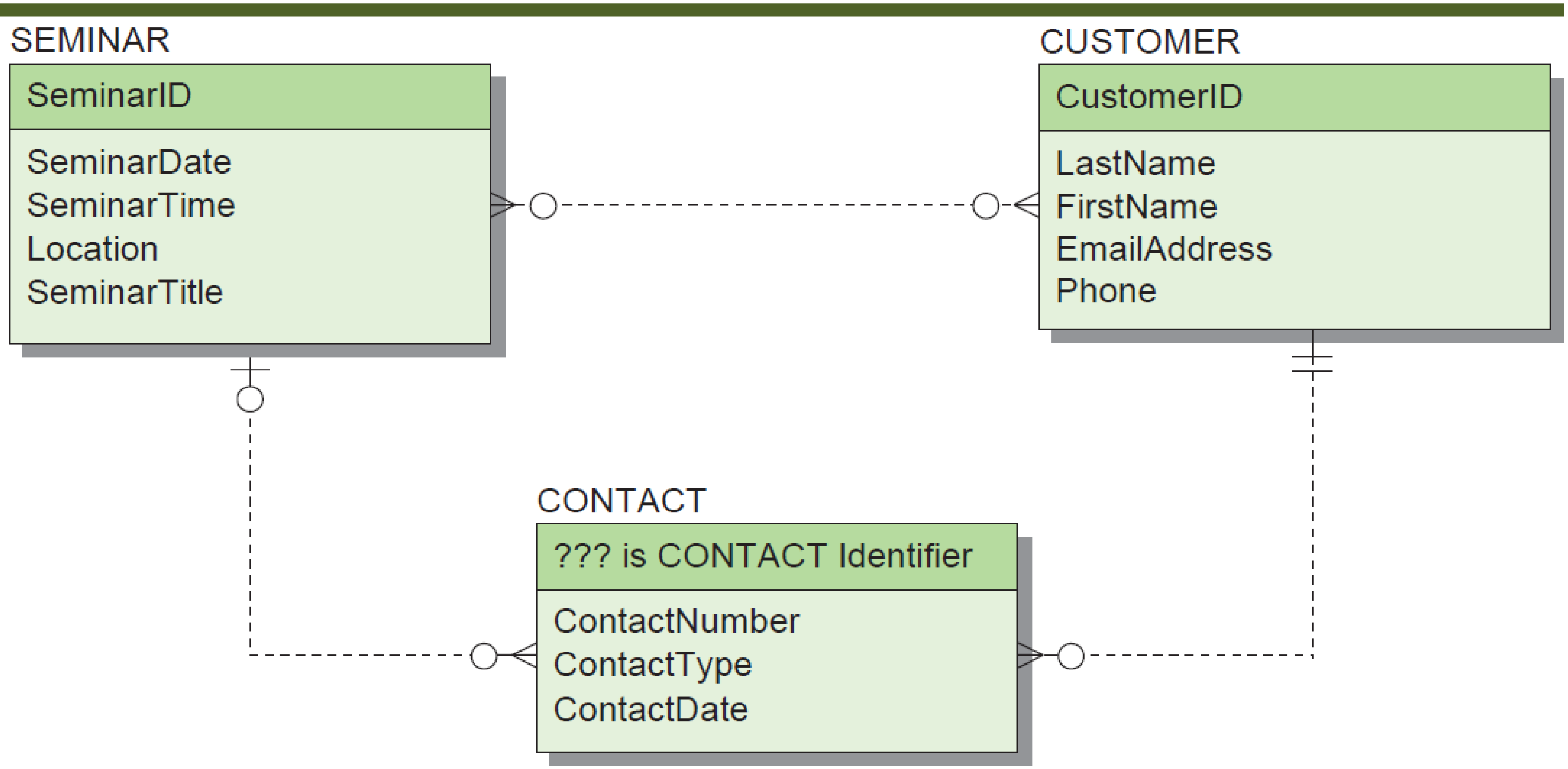
Thanks again and I look forward to hearing from you!

Best regards,

Heather Sweeney

Figure 4-18:
Heather Sweeney Designs Customer Form Letter

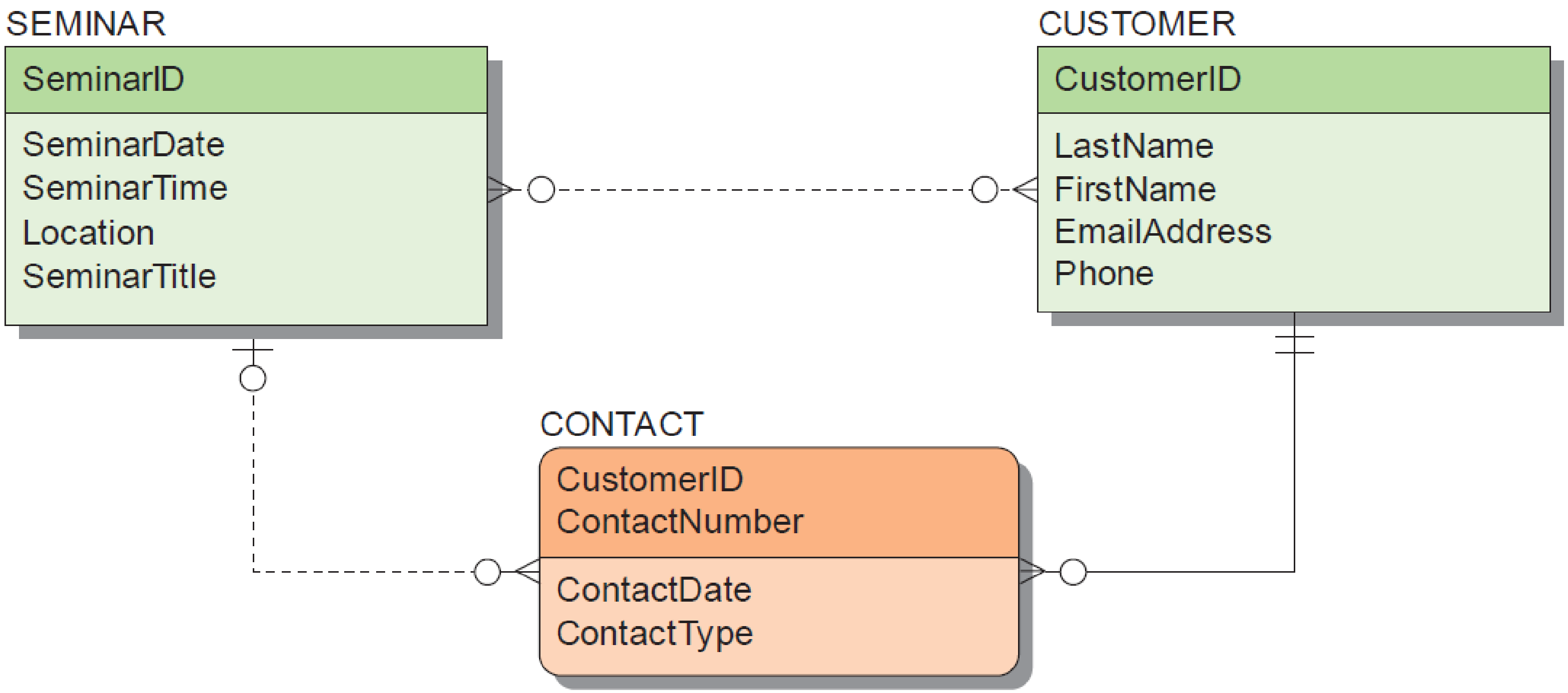
Heather Sweeney Designs: Data Model with CONTACT



(a) First Version with CONTACT

Figure 4-19: Heather Sweeney Designs Data Model with CONTACT

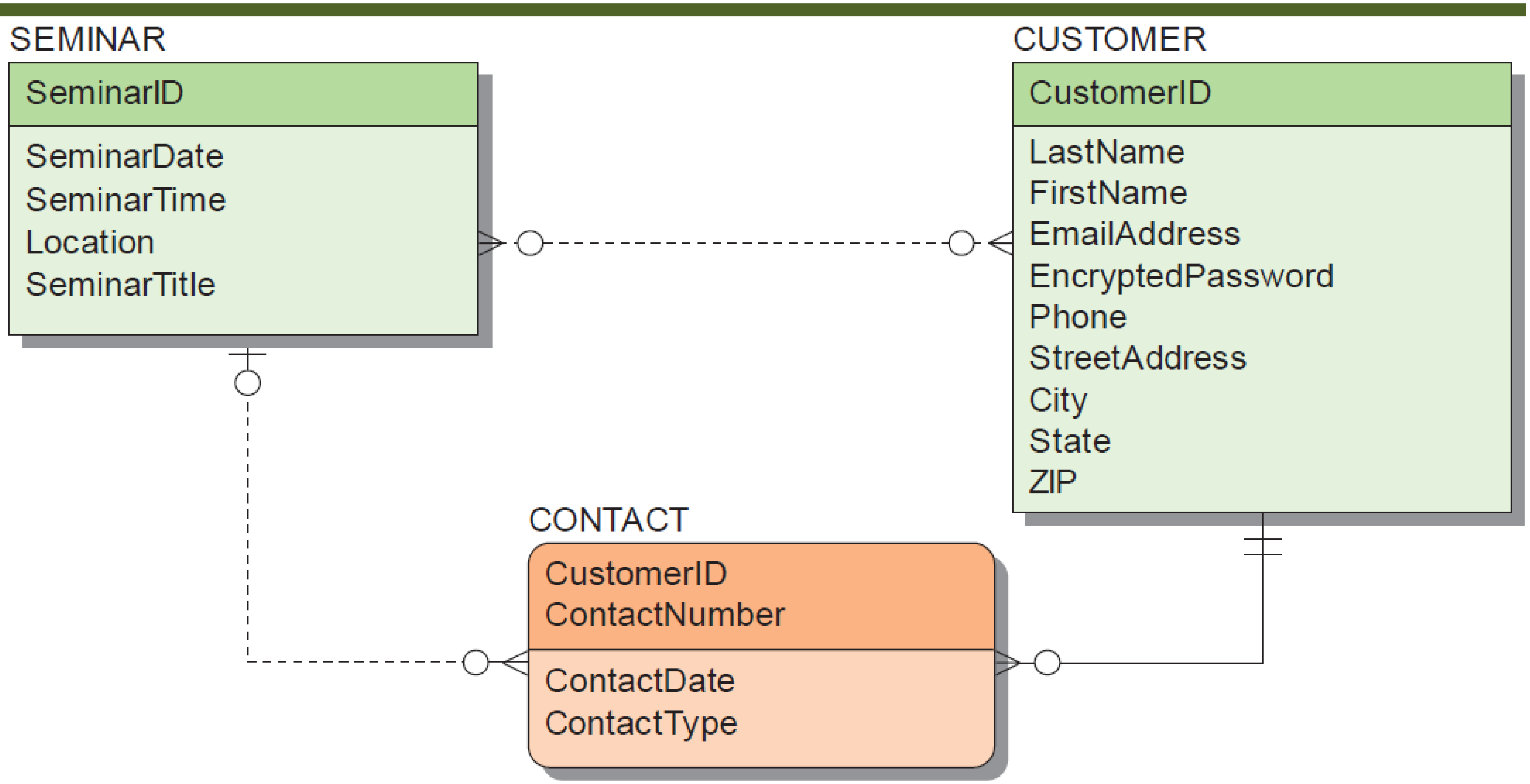
Heather Sweeney Designs: Data Model with CONTACT as Weak Entity



(b) Second Version with CONTACT as a Weak Entity

Figure 4-19: Heather Sweeney Designs Data Model with CONTACT

Heather Sweeney Designs: Data Model with Modified CUSTOMER



(c) Third Version with Modified CUSTOMER

Figure 4-19: Heather Sweeney Designs Data Model with CONTACT

Heather Sweeney Designs: Sales Invoice

Heather Sweeney Designs
122450 Rockaway Road
Dallas, Texas 75227

Invoice No. **35000**

INVOICE

Customer

Name Ralph Able
Address 123 Elm Street
City San Antonio State TX ZIP 78214
Phone 210-281-7987

Misc

Date 10/15/14
Order No.
Rep
FOB

Qty	Description	Unit Price	TOTAL
1	Kitchen Remodeling Basics - Video	$ 14.95	$ 14.95
1	Kitchen Remodeling Basics - Video Companion	$ 7.99	$ 7.99

	Subtotal	$ 22.94
	Shipping	$ 5.95
Tax Rate(s)	5.70%	$ 1.31
	TOTAL	$ 30.20

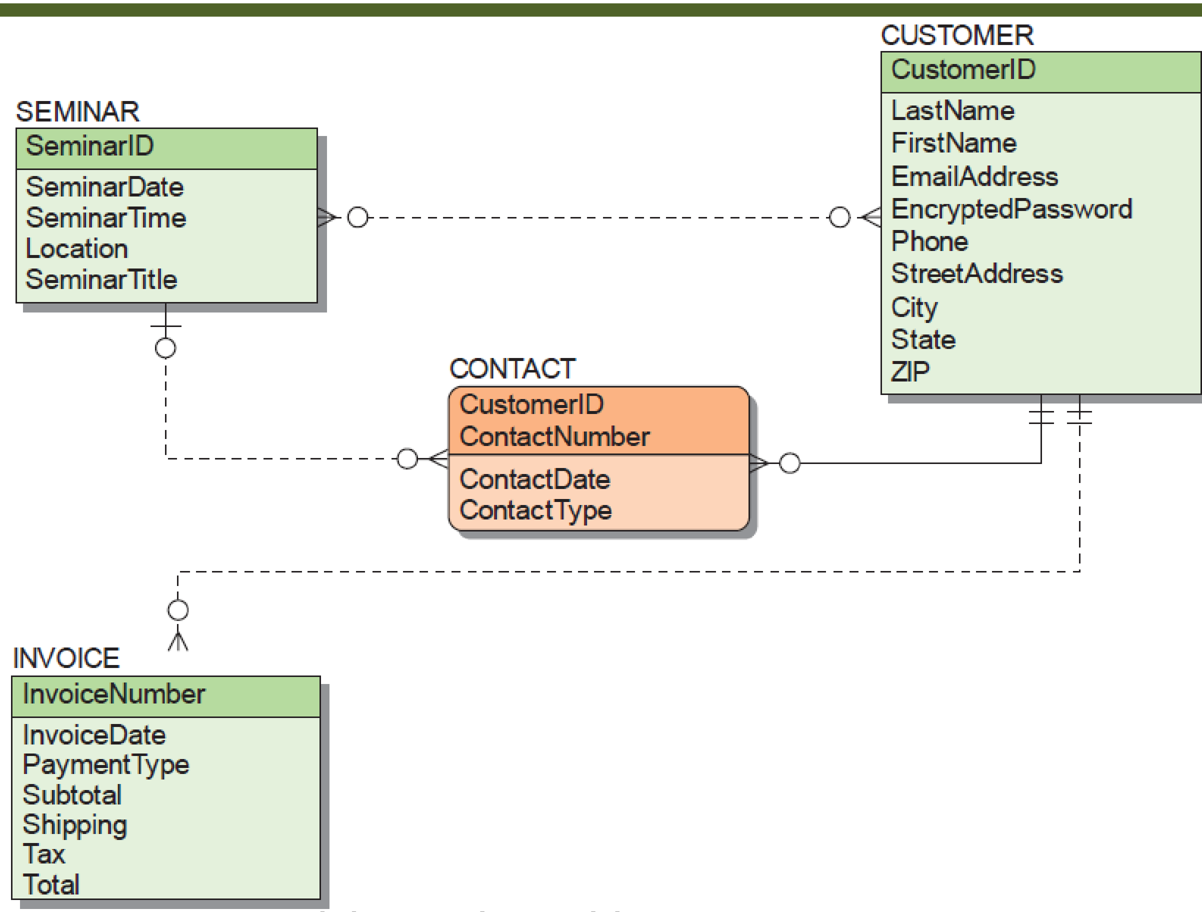
Payment Credit

Comments Visa
Name Ralph J. Able
CC # xxxx xxx xxx xxxxxx
Expires May-17

Office Use Only

Figure 4-20: Heather Sweeney Designs Sales Invoice

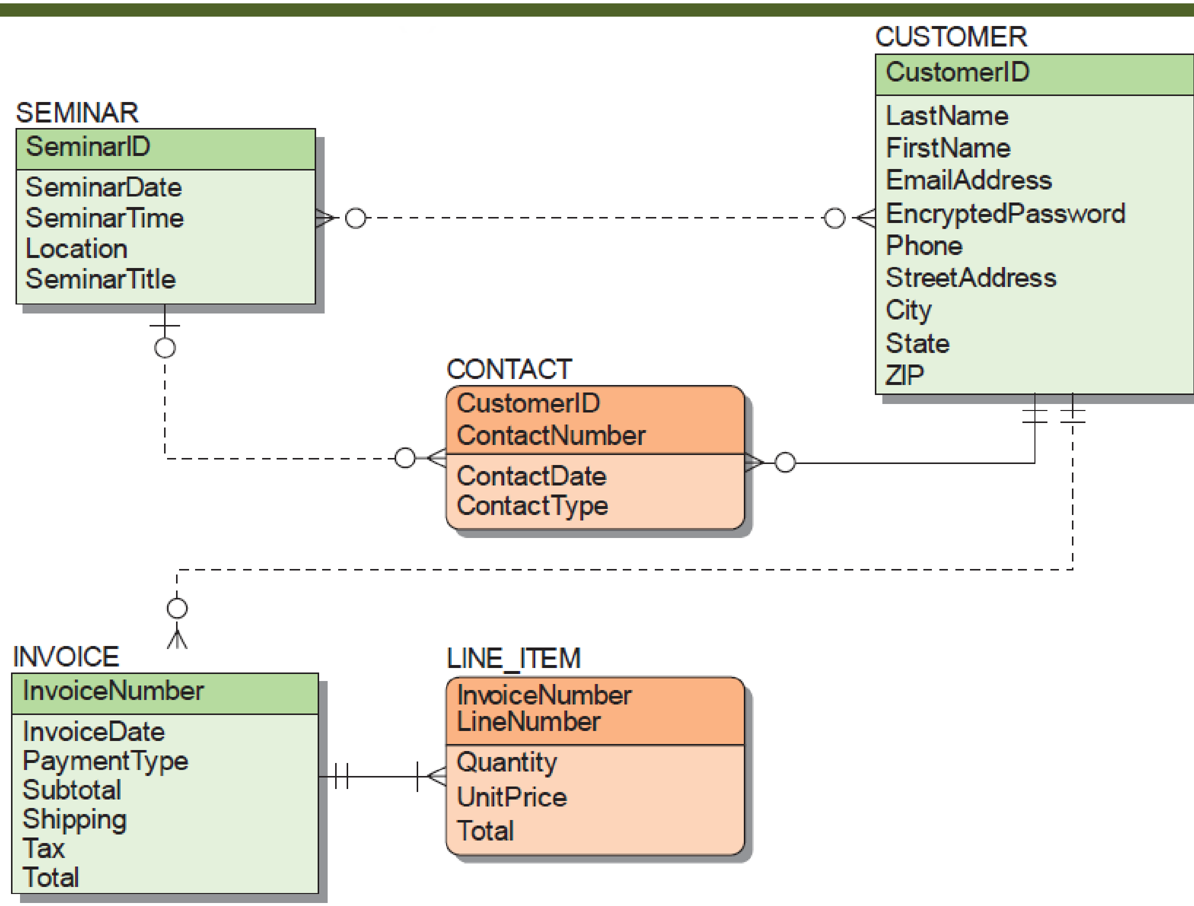
Heather Sweeney Designs: Data Model with INVOICE



(a) Version with INVOICE

Figure 4-21: The Final Data Model for Heather Sweeney Designs

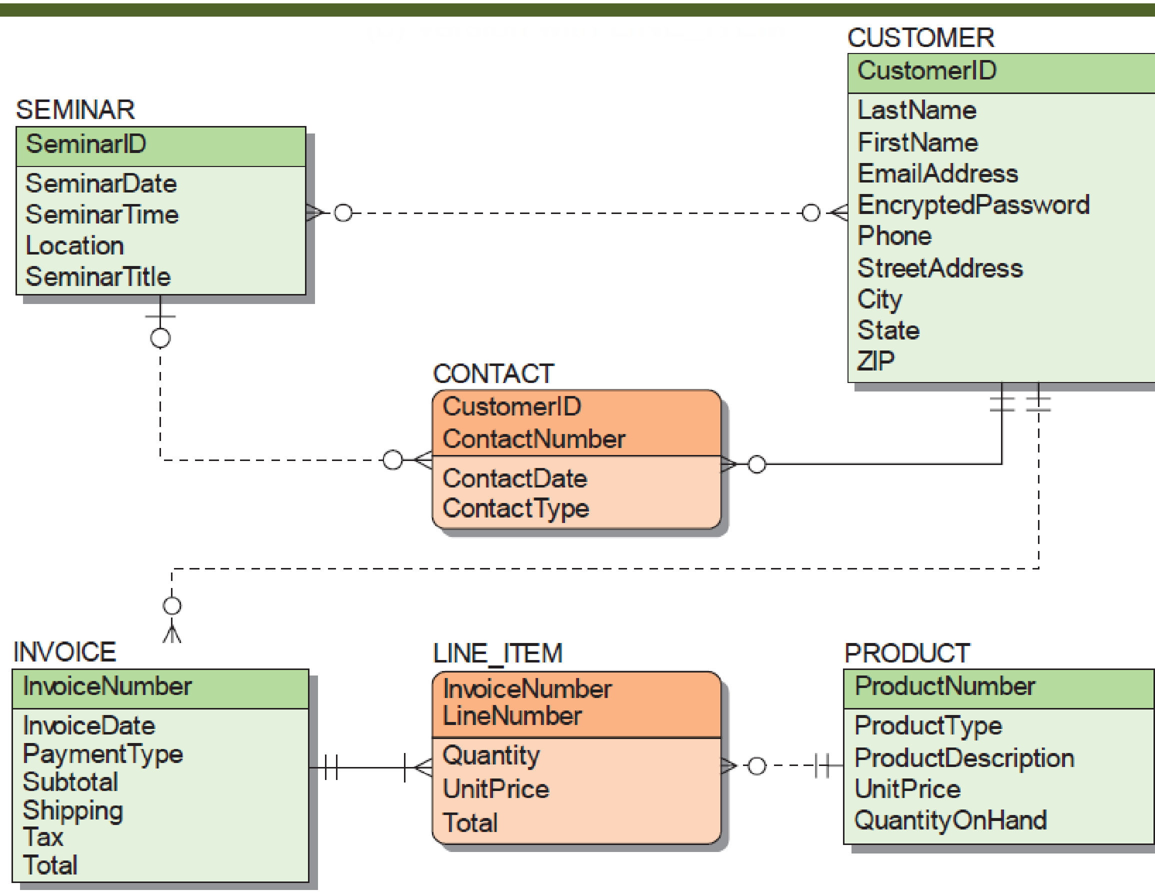
Heather Sweeney Designs: Data Model with LINE_ITEM



(b) Version with LINE_ITEM

Figure 4-21: The Final Data Model for Heather Sweeney Designs

Heather Sweeney Designs: Final Data Model



(c) The Finished Data Model

Figure 4-21: The Final Data Model for Heather Sweeney Designs

Heather Sweeney Designs: Business Rules and Model Validation

- **Business rules** may constrain the model and need to be recorded.
  - Heather Sweeney Designs has a business rule that no more than one form letter or email per day is to be sent to a customer.
- After the data model has been completed, it needs to be validated.
  - **Prototyping** is commonly used to validate forms and reports.

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End of Presentation on Chapter Four

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